

CBSE Class 12 Physics — Wave Optics

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQ/Assertion-Reason/VSA)

- Q1.** A ray of monochromatic light propagating in air is incident on the surface of water. Which of the following will be the exact same for both the reflected and refracted rays? (a) Energy carried (b) Speed (c) Frequency (d) Wavelength. [PYQ 2023] Very High
- Q2.** In a Young's double slit experiment, the path difference at a certain point on the screen between two interfering waves is $\frac{1}{8}^{\text{th}}$ of the wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to: (a) **0.80** (b) **0.74** (c) **0.94** (d) **0.85**. [PYQ 2022] Very High
- Q3.** The shape of the interference fringes in Young's double slit experiment when D (distance between slit and screen) is very large as compared to the fringe width is nearly: (a) straight line (b) parabolic (c) circular (d) hyperbolic. [PYQ 2020] High
- Q4. Assertion (A):** In Young's double slit experiment, all bright fringes are of equal width. **Reason (R):** The fringe width depends directly upon the wavelength of light (λ) used, the distance of the screen from the plane of slits (D), and inversely on the slit separation (d). [PYQ 2023] Very High
- Q5. Assertion (A):** In an interference pattern observed in Young's double slit experiment, if the separation (d) between coherent sources as well as the distance (D) of the screen from the coherent sources both are reduced to $\frac{1}{3}^{\text{rd}}$, then the new fringe width remains exactly the same. **Reason (R):** Fringe width is strictly proportional to (D/d) . [PYQ 2022] Very High
- Q6.** In Young's double slit experiment, the path difference between two interfering waves at a point on the screen is $5\lambda/2$, where λ is the wavelength of the light used. Which specific dark fringe will lie at this point? [PYQ 2020] High
- Q7.** If one of the slits in a Young's double slit experiment is fully closed, the new pattern on the screen has a __ central maximum in angular size compared to the original interference fringes. [PYQ 2020] High
- Q8.** Write the explicit condition on the path difference (Δx) under which destructive interference occurs in Young's double slit experiment. [PYQ 2020] Very High
- Q9.** When light travels from a rarer to a denser medium, its speed undeniably decreases. Does this reduction in speed imply a reduction in the energy carried by the light wave? [PYQ 2022] High
- Q10.** In the wave picture of light, the intensity of light is determined by the square of the amplitude of the wave. What specifically determines the intensity of light in the photon (quantum) picture of light? [PYQ 2022] Medium
- Q11.** What type of primary wavefront will emerge from a distant light source like a star? [PYQ 2022] Very High
- Q12.** What type of primary wavefront will emerge from an isotropic point source of light? [PYQ 2022] Very High
- Q13.** Monochromatic light of wavelength **589 nm** is incident from air on a water surface. What will be the exact wavelength of the reflected light? [PYQ 2022] Very High
- Q14.** In an interference pattern, the ratio of the amplitudes of the two interfering waves is **2 : 1**. What will be the ratio of the maximum intensity to the minimum intensity ($I_{\text{max}}/I_{\text{min}}$) in the fringe pattern? [PYQ 2015] High
- Q15.** Can two independent monochromatic sources (like two separate sodium lamps) produce a sustained interference pattern? [PYQ 2015] Very High
- Q16.** What is the phase difference between two coherent waves if the ratio of their resultant intensity at a point to the maximum possible intensity is **3 : 4**? [Practice] High
- Q17.** How is the angular width of the central maximum in a single slit diffraction pattern affected when the wavelength of the incident light is decreased? [PYQ 2022] High
- Q18.** Draw the shape of the emergent wavefront when a plane wavefront is incident on a thin convex lens. [PYQ 2016] High
- Q19.** Write the mathematical relation defining the refractive index (μ) of a medium in terms of the speed of light in a vacuum and in the medium. [PYQ 2019] Medium
- Q20.** Two coherent sources are essential for generating a stable and observable interference pattern. Define what makes two sources 'coherent'. [PYQ 2014] Very High

2-Mark Questions

- Q21.** Two slits are made **1 mm** apart and the screen is placed **1 m** away. What is the fringe separation when blue-green light of wavelength **500 nm** is used? [PYQ 2022] Very High
- Q22.** In a single slit diffraction experiment, the width of the slit is increased. How will the (A) size (angular width), and (B) intensity of the central bright band be affected? Justify your answer. [PYQ 2020] Very High
- Q23.** How will the interference pattern in Young's double-slit experiment be practically affected if (A) the screen is moved away from the plane of the slits, and (B) the monochromatic source is replaced by white light? [PYQ 2022] High

Q24. Draw the intensity distribution graphs side-by-side for (i) a single slit diffraction pattern, and (ii) a double slit interference pattern. [PYQ 2017] Very High

Q25. In a single slit diffraction pattern, why does the intensity of a secondary maximum become significantly less as compared to the central maximum? [PYQ 2022] High

Q26. In a Young's double-slit experiment, the distance between the slits is increased ten times whereas their distance from the screen is halved. What is the new fringe width in terms of the original fringe width β ? [PYQ 2022] High

Q27. If one of the two identical slits producing interference in Young's experiment is covered with glass so that the light intensity passing through it is reduced to 50%, find the exact ratio of the maximum and minimum intensity of the fringe in the interference pattern. [PYQ 2018] High

Q28. Two waves from two coherent sources S and S' superimpose at point X . If X is a point on the second minima and the path difference $SX - S'X$ is 4.5 cm, calculate the wavelength of the waves. [PYQ 2021] Medium

Q29. A monochromatic light produces a diffraction pattern through a slit of width 1 mm on a screen that is 2 m away. If the linear width of the central maxima in the diffraction pattern is 2 cm, determine the wavelength of the incident light. [PYQ 2025-26 SQP] High

Q30. Briefly explain the conditions required for two light waves to interfere constructively and destructively at a point, in terms of their phase difference ($\Delta\phi$). [PYQ 2020] Very High

Q31. State Huygens' principle of secondary wavelets. [PYQ 2016] Very High

Q32. Does the appearance of bright and dark fringes in an interference pattern violate, in any way, the law of conservation of energy? Explain briefly. [PYQ 2015] High

3-Mark Questions

Q33. A beam of light consisting of two wavelengths 600 nm and 480 nm is used in a Young's double slit experiment. The slit separation is 0.6 mm and the screen is kept 1.0 m away from the plane of the slits. Find: (A) The distance of the second bright fringe from the central maximum pertaining to the light of 600 nm. (B) The least distance from the central maximum at which bright fringes due to both wavelengths coincide. [PYQ 2022] Very High

Q34. A beam of light consisting of two wavelengths 600 nm and 500 nm is used in a YDSE. The slit separation is 1.0 mm and the screen is kept 0.60 m away. Calculate the least distance from the central maximum where the bright fringes due to both the wavelengths coincide. [PYQ 2022] High

Q35. Define a wavefront. Using Huygens' construction of secondary wavelets, draw a diagram showing the passage of a plane wavefront from a denser into a rarer medium, and hence verify Snell's law of refraction. [PYQ 2022, 2015] Very High

Q36. State Huygens' principle. With the help of a suitable diagram, show how a plane wave is reflected from a surface. Hence verify the law of reflection. [PYQ 2023, 2019] Very High

Q37. Describe any three characteristic features which distinguish an interference pattern in Young's double slit experiment from a single slit diffraction pattern. [PYQ 2019, 2016] Very High

Q38. The ratio of the intensities at minima to the maxima in a Young's double slit experiment is 9 : 25. Find the ratio of the widths of the two slits. [PYQ 2014] High

Q39. A narrow slit is illuminated by a parallel beam of monochromatic light of wavelength $\lambda = 6000 \text{ \AA}$ and the angular width of the central maximum in the resulting diffraction pattern is measured. When the slit is next illuminated by light of an unknown wavelength λ' , the angular width strictly decreases by 30%. Calculate the exact value of the wavelength λ' . [PYQ 2022] Medium

Q40. A coherent light beam of 500 nm wavelength strikes two slits, producing an interference pattern. The two slits are separated by 0.3 mm and the distance between the slits and the screen is 0.3 m. Determine the total number of maxima observed across the perpendicular distance from $y = -1 \text{ mm}$ to $y = +1 \text{ mm}$ on the screen. [PYQ 2025-26 SQP] High

Q41. A source of light emitting two wavelengths, 5890 \AA and 5895 \AA, is used in a Young's double slit experiment. Due to the dual wavelengths, the fringe patterns overlap on the screen. Determine up to what order (n) the fringes can be seen clearly before the pattern appears continuous. [PYQ 2025-26 SQP] High

Q42. What should be the specific width of a single slit to obtain n maxima of a double slit pattern (with slit separation d) within the central maxima of the single slit pattern? [PYQ 2021] Medium

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Define a wavefront. How is it fundamentally different from a ray? (B) Depict the shape of a wavefront in each of the following cases: (i) Light diverging from a point source. (ii) Light emerging out of a convex lens when a point source is placed exactly at its focus. (C) What will be the shape of the wavefront intercepted by a large reflecting type telescope on earth, due to a star far-away from our solar system? (D) Use Huygens' principle to prove the laws of reflection. [PYQ 2015, 2023, 2025-26 SQP] Very High

Q44. (A) Why cannot the phenomenon of interference be observed by illuminating two separate pinholes with two identical sodium lamps? (B) Two monochromatic waves having displacements $y_1 = a \cos \omega t$ and $y_2 = a \cos(\omega t + \phi)$ from two

coherent sources interfere to produce an interference pattern. Derive the mathematical expression for the resultant intensity and obtain the phase conditions for constructive and destructive interference. (C) In a YDSE, what kind of fringes do you expect to observe if white light is used instead of monochromatic light? [PYQ 2019, 2018] Very High

Q45. (A) Explain the phenomenon of diffraction of light at a single slit to show the formation of fractional fringes. (B) Show graphically the angular distribution of intensity in a single slit diffraction pattern. (C) For a single slit of width a , the first minimum of the interference pattern of a monochromatic light of wavelength λ occurs at an angle of λ/a . However, at the exact same angle of λ/a , we get a bright maximum for two narrow slits separated by a distance a . Explain this apparent contradiction. [PYQ 2014, 2022] High

Q46. (A) In Young's double slit experiment, deduce the path difference condition for (i) constructive, and (ii) destructive interference at a point on the screen. (B) Draw a proper graph showing the variation of intensity in the interference pattern against position ' x ' on the screen. (C) Does the interference phenomenon strictly obey the principle of conservation of energy? Give a mathematical justification. [PYQ 2016, 2015] Very High

Q47. (A) State the essential conditions for two light waves to be coherent. (B) In a double slit experiment, given that $d = 0.140 \text{ mm}$, $D = 140 \text{ cm}$, and $\lambda = 600 \text{ nm}$. Find the position of the 3rd bright fringe on the screen. (C) A monochromatic light produces a fringe pattern on the screen in a YDSE setup. As the screen is moved towards the slits through a distance of 0.05 m , the change in the fringe width is observed to be $3 \times 10^{-5} \text{ m}$. If the separation between the slits is 10^{-3} m , determine the wavelength of the light used. [PYQ 2025-26 SQP, 2022] High

Q48. (A) Describe the exact geometrical shape of the wavefronts for (i) a point source, (ii) a line source, and (iii) a distant source. (B) How will the interference pattern in a YDSE change if the phase difference between the light waves emanating from the two slits changes from 0 to π and then remains constant? (C) Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6} \text{ m}$. If the distance between the slit and the screen is 1.5 m , calculate the separation between the physical positions of the second maxima of the diffraction pattern obtained in the two individual cases. [PYQ 2022, 2019] High

Q49. (A) In a YDSE, the source slit is moved slightly away from the plane of the double slits. How does this specifically affect the interference pattern? (B) A blue coloured alternate band of diffraction pattern appears on a screen due to the blue light of wavelength λ passing through a single slit of width w . If the blue light is replaced with a green light of wavelength 1.5λ , to what exact width should you change the slit in order to get the original angular pattern back? (C) At what distance should a screen be placed from the slits ($d = 0.4 \text{ mm}$) in a YDSE illuminated by $\lambda = 500 \text{ nm}$, in order that a dark fringe appears directly opposite to both the slits and only one bright fringe exists strictly in between them? [PYQ 2025-26 SQP, 2022] Medium

Q50. (A) Write the expression for the fringe width in Young's double-slit experiment. (No derivation). (B) Based on this expression, state three parameters on which the fringe width depends, and how it varies with them. (C) If the entire apparatus of a Young's double-slit experiment is immersed in water (refractive index $4/3$), how does the fringe width change? Prove it mathematically. (D) Sea waves of wavelength 25 m keep splashing at a 60 m wide opening in a wall barrier. Racing boats are parked on the other side. Determine the first angle to the incident direction at which the boats can be parked so that they experience minimum rocking motion (first diffraction minimum). [Practice, PYQ 2025-26 SQP] High

Answer Key

Q1: (c) Frequency. (Frequency depends only on the source, not the medium). **Q2:** (d) 0.85 . (

$\Delta x = \lambda/8 \Rightarrow \Delta \phi = \frac{2\pi}{\lambda}(\lambda/8) = \pi/4$. Intensity $I = I_0 \cos^2(\frac{\Delta \phi}{2}) = I_0 \cos^2(\pi/8) \approx 0.85I_0$. Ratio is 0.85). **Q3:** (a) straight line. (They are hyperbolic near the center but appear as straight lines for very large D). **Q4:** (a) Both A and R are true and R is the correct explanation. ($\beta = \lambda D/d$). **Q5:** (a) Both A and R are true and R is the correct explanation. (

$\beta' = \lambda(D/3)/(d/3) = \lambda D/d = \beta$). **Q6:** The 3rd dark fringe. (Dark fringe condition: $\Delta x = (2n - 1)\lambda/2$.

$5\lambda/2 = (2n - 1)\lambda/2 \Rightarrow 2n - 1 = 5 \Rightarrow n = 3$). **Q7:** Broader (or wider). (It becomes a single-slit diffraction pattern, where the central maximum has an angular width of $2\lambda/a$, wider than interference fringes). **Q8:** $\Delta x = (2n - 1)\frac{\lambda}{2}$, where $n = 1, 2, 3, \dots$ **Q9:** No. The energy of a light wave is given by $E = h\nu$. Since frequency ν remains unchanged during refraction, energy remains constant. **Q10:** The number of photons crossing a unit area per unit time. **Q11:** Plane wavefront.

Q12: Spherical wavefront. **Q13:** 589 nm . (Wavelength does not change during reflection in the same medium). **Q14:** $9 : 1$. ($I \propto A^2 \Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(2+1)^2}{(2-1)^2} = \frac{9}{1}$). **Q15:** No. They emit light waves with a rapidly varying, random phase difference, making them incoherent. **Q16:** $\pi/3$ or 60° . (

$I = I_{\max} \cos^2(\phi/2) \Rightarrow 3/4 = \cos^2(\phi/2) \Rightarrow \cos(\phi/2) = \sqrt{3}/2 \Rightarrow \phi/2 = \pi/6 \Rightarrow \phi = \pi/3$). **Q17:** It decreases. (Angular width $\theta = 2\lambda/a$, so $\theta \propto \lambda$). **Q18:** Plane wavefront. (Spherical wavefront from focus becomes parallel rays, which represent a plane wavefront). **Q19:** $\mu = c/v$. **Q20:** Two sources are coherent if they emit light waves of the same frequency and maintain a constant phase difference with time.

Q21: $\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{1 \times 10^{-3}} = 5 \times 10^{-4} \text{ m} = 0.5 \text{ mm}$. **Q22:** (A) Size decreases (Angular width $= 2\lambda/a$, so it is inversely proportional to width a). (B) Intensity increases (More light energy passes through a wider slit). **Q23:** (A) Fringe width β increases ($\beta \propto D$). (B) The central fringe will be white, flanked by a few coloured fringes (violet closest, red farthest), after which the pattern becomes uniform illumination. **Q24:** Draw a uniform sinusoidal squared curve for interference. For diffraction, draw a prominent central peak flanked by rapidly diminishing secondary peaks. **Q25:** In the central maximum,

wavelets from all parts of the slit arrive in phase. For secondary maxima, wavelets from only a small fraction of the slit (e.g., $1/3$ for 1st max, $1/5$ for 2nd max) interfere constructively, while the rest cancel out. **Q26:** $\beta' = \frac{\lambda(D/2)}{10d} = \frac{1}{20} \frac{\lambda D}{d} = \frac{\beta}{20}$. **Q27:** Let $I_1 = I_0$, $I_2 = 0.5I_0$.

$I_{\max}/I_{\min} = (\sqrt{I_1} + \sqrt{I_2})^2/(\sqrt{I_1} - \sqrt{I_2})^2 = (1 + \sqrt{0.5})^2/(1 - \sqrt{0.5})^2 = (1 + 0.707)^2/(1 - 0.707)^2 \approx (1.707/0.293)^2 \approx 34$. **Q28:** 2nd minima condition: $\Delta x = 3\lambda/2$. $4.5 = 3\lambda/2 \Rightarrow \lambda = 3 \text{ cm}$. **Q29:** Linear width $= 2\lambda D/a$.

$2 \times 10^{-2} = 2 \times \lambda \times 2/(1 \times 10^{-3}) \Rightarrow \lambda = \frac{2 \times 10^{-5}}{4} = 0.5 \times 10^{-5} \text{ m} = 5000 \text{ \AA}$ (or 500 nm). **Q30:** Constructive:

$\Delta\phi = 2n\pi$ (even multiple of π). Destructive: $\Delta\phi = (2n - 1)\pi$ (odd multiple of π). **Q31:** Every point on a given wavefront acts as a fresh source of secondary wavelets, which travel in all directions with the speed of light. The forward envelope of these wavelets at any instant gives the new wavefront. **Q32:** No. Energy is merely redistributed. The energy missing at dark fringes appears exactly at the bright fringes. Average intensity remains $I_1 + I_2$, perfectly conserving energy.

Q33: (A) $y = n\lambda_1 D/d = 2 \times 600 \times 10^{-9} \times 1/(0.6 \times 10^{-3}) = 2 \text{ mm}$. (B) Condition for coincidence:

$n\lambda_1 = (n + 1)\lambda_2 \Rightarrow 600n = 480(n + 1) \Rightarrow 120n = 480 \Rightarrow n = 4$. Least distance

$y = 4 \times 600 \times 10^{-9} \times 1/(0.6 \times 10^{-3}) = 4 \text{ mm}$. **Q34:**

$n(600) = (n + 1)(500) \Rightarrow 600n = 500n + 500 \Rightarrow 100n = 500 \Rightarrow n = 5$. Distance

$y = 5 \times 600 \times 10^{-9} \times 0.6/10^{-3} = 1.8 \times 10^{-3} \text{ m} = 1.8 \text{ mm}$. **Q35:** Draw incident plane wavefront AB . Let speed in denser be v_1 , rarer be v_2 . By the time B reaches interface at C ($t = BC/v_1$), secondary wavelet from A travels $AD = v_2 t$.

CD is the refracted wavefront. $\sin i / \sin r = (BC/AC)/(AD/AC) = BC/AD = v_1/v_2 = \mu_2/\mu_1$ (Snell's Law). **Q36:** Draw incident wavefront AB on reflecting surface. Let B hit surface at C in time t . Radius of wavelet from A is $AD = ct$. Since $AD = BC$ and AC is common, $\triangle ADC \cong \triangle CBA$. Thus $\angle i = \angle r$. **Q37:** 1. Interference is due to superposition of waves from two distinct coherent sources; diffraction is due to superposition of secondary wavelets from different parts of the same wavefront. 2. Interference fringes are generally of equal width; diffraction fringes are not (central max is double the width). 3. All bright interference fringes have same intensity; diffraction maxima intensity falls off rapidly. **Q38:**

$I_{\min}/I_{\max} = 9/25 \Rightarrow (a_1 - a_2)^2/(a_1 + a_2)^2 = 9/25 \Rightarrow (a_1 - a_2)/(a_1 + a_2) = 3/5 \Rightarrow 5a_1 - 5a_2 = 3a_1 + 3a_2 \Rightarrow 2a_1 = 8a_2$. Width ratio $w_1/w_2 = I_1/I_2 = a_1^2/a_2^2 = 16/1$. **Q39:** Angular width $\theta \propto \lambda$. A 30% decrease means new width is 70% of original. So $\lambda' = 0.70 \times 6000 = 4200 \text{ \AA}$. **Q40:** Maxima at

$y = n\lambda D/d = n(500 \times 10^{-9})(0.3)/(0.3 \times 10^{-3}) = n \times 0.5 \text{ mm}$. For $-1 \text{ mm} \leq y \leq 1 \text{ mm}$, n can be $-2, -1, 0, 1, 2$. Total = 5 maxima. **Q41:** Overlap becomes continuous when n^{th} maximum of λ_1 coincides with $(n - 1)^{\text{th}}$ minimum (or essentially the phase difference washes out). By standard formula for continuous overlap limit:

$n\lambda_1 = (n + 1/2)\lambda_2 \Rightarrow n(5890) = (n + 0.5)5895 \Rightarrow 5890n = 5895n + 2947.5 \Rightarrow -5n = 2947.5$. Wait, $n\lambda_2 = (n + 1/2)\lambda_1 \Rightarrow 5895n = 5890n + 2945 \Rightarrow 5n = 2945 \Rightarrow n = 589$. Fringes are seen up to order 589. **Q42:** Width of single slit central maximum $= 2\lambda/a$. Width of n double slit fringes $= n(\lambda/d)$. Equating: $n\lambda/d = 2\lambda/a \Rightarrow a = 2d/n$.

Q43: (A) A continuous locus of particles vibrating in the same phase. A ray is an imaginary line perpendicular to the wavefront pointing in the direction of energy propagation. (B) (i) Spherical, (ii) Plane. (C) Plane wavefront. (D) Draw diagram with incident plane wavefront AB and reflected wavefront CD . Prove $\triangle BAC \cong \triangle DCA \Rightarrow \angle i = \angle r$. **Q44:** (A) Independent sources undergo rapid and random phase changes, resulting in a time-averaged interference term of zero, producing uniform illumination. (B) $y = y_1 + y_2 = a[\cos \omega t + \cos(\omega t + \phi)] = 2a \cos(\phi/2) \cos(\omega t + \phi/2)$. Intensity $I \propto \text{Amplitude}^2 = 4a^2 \cos^2(\phi/2)$. Max when $\phi = 2n\pi$, Min when $\phi = (2n - 1)\pi$. (C) Central fringe is white. Fringes closest to center are violet/blue; farthest are red. Higher orders wash out into uniform illumination. **Q45:** (A) Divide the slit into n equal parts. If path difference between edges is $n\lambda$, every point in upper half of a pair cancels a point in the lower half, causing a minimum. (B) Draw a central peak of width $2\lambda/a$, with secondary peaks dropping heavily in intensity (like $1/22$, $1/61$). (C) For a single slit of width a , wavelets from top and middle cancel ($\Delta x = \lambda/2$), yielding minimum. For two distinct slits separated by a , $\Delta x = a \sin \theta = \lambda$, yielding constructive interference (max). **Q46:** (A) Path diff $\Delta x = yd/D$. (i)

Constructive: $\Delta x = n\lambda$. (ii) Destructive: $\Delta x = (2n - 1)\lambda/2$. (B) Draw perfectly uniform sinusoidal intensity peaks vs x . (C) Yes. The energy missing at destructive minima is simply relocated to the constructive maxima. The average intensity across the pattern is $2I_0$, which exactly equals the sum of individual intensities $I_0 + I_0$. **Q47:** (A) Must emit waves of same frequency and constant phase difference. (B)

$y_3 = 3\lambda D/d = 3(600 \times 10^{-9})(1.4)/(0.14 \times 10^{-3}) = 1.8 \times 10^{-2} \text{ m} = 1.8 \text{ cm}$. (C)

$\beta = \lambda D/d \Rightarrow \Delta\beta = \lambda \Delta D/d \Rightarrow 3 \times 10^{-5} = \lambda(0.05)/10^{-3} \Rightarrow \lambda = 3 \times 10^{-8}/0.05 = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$. **Q48:** (A)

(i) Spherical, (ii) Cylindrical, (iii) Plane. (B) The entire fringe pattern shifts by half a fringe width. The original central bright fringe becomes a completely dark fringe. (C) Second max condition: $\theta = 5\lambda/2a$. Linear pos $y = 5\lambda D/2a$.

$\Delta y = 5(\lambda_2 - \lambda_1)D/2a = 5(596 - 590) \times 10^{-9} \times 1.5/(2 \times 2 \times 10^{-6}) = 5 \times 6 \times 10^{-9} \times 1.5/(4 \times 10^{-6}) = 11.25 \times 10^{-3} \text{ m}$.

Q49: (A) It reduces the spatial coherence of the light hitting the double slits. If moved too far, the interference fringes lose sharpness and eventually disappear. (B) Original pattern angle $\theta = \lambda/w$. New angle $\theta = 1.5\lambda/w'$. For same pattern, $\lambda/w = 1.5\lambda/w' \Rightarrow w' = 1.5w$. (C) First dark fringes at $y = \pm \lambda D/2d$. Distance between them is the single bright fringe.

The slits are at $y = \pm d/2$. For dark fringe to be directly opposite to the slits,

$\lambda D/2d = d/2 \Rightarrow D = d^2/\lambda = (0.4 \times 10^{-3})^2/(500 \times 10^{-9}) = 0.16 \times 10^{-6}/(500 \times 10^{-9}) = 0.32 \text{ m} = 32 \text{ cm}$. **Q50:**

(A) $\beta = \lambda D/d$. (B) Directly proportional to wavelength (λ), directly proportional to screen distance (D), inversely proportional to slit separation (d). (C) In water, wavelength decreases to $\lambda' = \lambda/\mu = 3\lambda/4$. Therefore, $\beta' = \beta/\mu = 3\beta/4$ (fringes become narrower). (D) Single slit first minimum:

$a \sin \theta = \lambda \Rightarrow 60 \sin \theta = 25 \Rightarrow \sin \theta = 25/60 \approx 0.416 \Rightarrow \theta = \sin^{-1}(0.416) \approx 24.6^\circ$.